SCHOOL OF COMPUTING
DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
Academic year 2025-2026
(Winter Semester)

Course code: 10214AD701 **Course Name: Major Project**

TTS NO	Supervisor Name	VTU NO.	Student Name1	VTU NO.	Student Name 2
	Dr.R.ANANDH	21352	SHYAAM.R.K	23339	NANDHA KUMAR.T

Title of the Project: **AI-Powered Predictive Analytics Framework for Early Diagnosis and Progression Monitoring of Varicose Vein Disorders**

Problem Statement

Current diagnostic methods for Varicose Vein Disorder (VVD) are subjective and reactive, often identifying the condition only after severe complications arise. To address this, the proposed AI framework integrates Doppler ultrasound, Electronic Health Records, and real-time wearable data (1 Hz sampling) into a hybrid CNN-LSTM model. This system utilizes Convolutional Neural Networks for image-based biomarker extraction and Long Short-Term Memory networks to forecast disease progression. By incorporating Explainable AI (XAI) such as SHAP and LIME, the model provides transparent, evidence-based insights—achieving 92.3% accuracy for early diagnosis and 89.5% for longitudinal progression monitoring.

Mapping of the Project to Complex Engineering Problem (CEP)

Level	Attribute	Characteristics	Targeted WKs (wherever applicable)	Justification (as applicable)
WP1	Depth of Knowledge	Cannot be resolved without in-depth engineering knowledge at the level of K3, K4, K5, K6 or K8	WK3, WK4, WK5, WK6, WK8	Requires deep knowledge of Deep Learning architectures (CNNs, LSTMs), signal processing for wearable data at 1 Hz, and specialized medical imaging (Doppler ultrasound) to extract biomarkers like reflux velocity.

WP2	Range of conflicting requirements	Involve wide-ranging and/or conflicting technical and non-technical issues	WK4, WK5, WK6, WK8	Addresses the conflict between high-accuracy "black box" models and the medical need for transparency/XAI. It also balances real-time computation on edge devices versus the computational intensity of deep CNNs .
WP3	Depth of analysis required	Have no obvious solution and require originality in analysis	WK3, WK4, WK5, WK8	No standard solution exists for predicting VVD progression. The project uses original hybrid modeling (CNN-LSTM) and feature-level data fusion to synchronize heterogeneous clinical, imaging, and wearable streams.
WP4	Familiarity of issues	Involve infrequently encountered or novel problems	WK4, WK8	The integration of continuous behavioral monitoring (posture/steps) via wearables with longitudinal ultrasound analysis to forecast future disease stages is a novel approach in vascular medicine.
WP5	Extent of applicable codes	Address problems not encompassed by standards and codes of practice	WK6, WK8	Current VVD diagnosis is subjective and operator-dependent. Standard clinical codes do not cover automated XAI-driven risk curves or the synchronization of real-time wearable signals with hospital PACS/EHR systems.
WP6	Stakeholder involvement and conflicting requirements	Involve collaboration across disciplines and diverse stakeholders	WK5, WK6, WK7	Requires collaboration between AI researchers, vascular specialists, and regulatory authorities to ensure the system meets clinical needs while maintaining patient confidentiality and ethical standards.

WP7	Interdependence	Address high-level problems requiring a systems approach	WK3, WK5, WK6	The framework is a modular multi-phase pipeline involving interdependent modules: Data Acquisition, Image Analysis, Clinical/Wearable Fusion, Predictive Modeling, XAI, and Deployment.

Mapping of the Project to Complex Engineering Activities (CEA)

EA No.	Attribute	Characteristics	Justification based on the Project
EA1	Range of Resources	Diverse resources and technologies	The project integrates a wide array of high-tech resources, including Doppler ultrasound imaging (B-mode and color), wearable posture and mobility sensors (sampling at 1 Hz), Electronic Health Records (EHR), and high-performance computing hardware such as the NVIDIA RTX 4090 GPU.
EA2	Level of Interactions	Optimization of complex interactions	It requires optimizing the interactions between heterogeneous data streams (imaging vs. temporal sensors), the synchronization of real-time wearable signals with hospital PACS and EHR systems, and the balance between deep learning model complexity and real-time inference latency.
EA3	Innovation	Creative and research-based solutions	The framework demonstrates innovation through a hybrid CNN-LSTM architecture that combines spatial imaging analysis with temporal behavioral tracking, further enhanced by Explainable AI (XAI) tools like SHAP, LIME, and Grad-CAM to ensure clinical transparency.
EA4	Consequences to Society and Environment	Significant societal impact	The system addresses a condition affecting 30% of the global adult population, aiming to prevent painful complications like venous ulcers and thrombosis through early detection, which reduces long-term healthcare costs and improves patient quality of life.
EA5	Familiarity	Extends beyond prior experience	This project extends beyond routine clinical practice by moving from subjective, reactive diagnosis to a proactive predictive model, utilizing emerging technologies like Edge AI and multimodal data fusion that are not yet standard in vascular medicine.

Sustainable Development Goals (SDG) Mapping

SDG No.	SDG Title	Relevance to the Project
SDG 9	Industry, Innovation and Infrastructure	Introduces an innovative hybrid AI architecture (CNN-LSTM) and cloud-integrated infrastructure that synchronizes real-time wearable signals with hospital EHR and PACS systems.
SDG 16	Peace, Justice and Strong Institutions	Strengthens clinical trust and ethical AI use through Explainable AI (XAI) models like SHAP and LIME, ensuring that medical decisions are transparent, evidence-based, and accountable.
SDG 10	Reduced Inequalities	Addresses healthcare inequality by supporting deployment on edge AI devices and portable ultrasound machines, making advanced diagnostic tools accessible in resource-constrained or hard-to-reach locations.
SDG 3	Good Health and Well-being	Directly improves the quality of care for patients with Varicose Vein Disorder (VVD) by enabling early diagnosis with 92.3% accuracy and proactive progression monitoring. This prevents severe complications such as venous ulcers and thrombosis.

Overall Justification: The project satisfies the criteria for Complex Engineering Problems (CEP) and Complex Engineering Activities (CEA) by transitioning Varicose Vein Disorder (VVD) diagnosis from subjective clinical methods to a proactive, AI-driven predictive framework. It requires multidisciplinary expertise in deep learning (CNNs and LSTMs), signal processing for 1 Hz wearable data, and vascular medicine to integrate heterogeneous data streams for longitudinal progression monitoring. By incorporating Explainable AI (XAI) tools like SHAP and LIME, the system resolves the conflict between high-accuracy "black box" modeling and the clinical necessity for transparent, evidence-based decision-making. The framework demonstrates significant societal impact by targeting a condition affecting 30% of the global population, offering a scalable, cloud-integrated solution that enables early intervention and reduces the long-term healthcare burden of advanced venous disease.

Depth of Knowledge – Required Knowledge Levels (K3–K8)

The problem addressed in this project cannot be resolved without in-depth engineering knowledge spanning the following Knowledge Profile levels:

K3 – Engineering Fundamentals

- Understanding of deep learning architectures, signal processing for temporal data, and medical database fundamentals for storing Electronic Health Records (EHR).
- Application of feature-level data fusion and data flow models to synchronize heterogeneous clinical, imaging, and wearable sensor streams.

K4 – Specialist Knowledge

- Advanced knowledge in Convolutional Neural Networks (CNNs) for medical image segmentation and Long Short-Term Memory (LSTM) networks for disease progression modeling.

- Expertise in Explainable AI (XAI) techniques, including SHAP, LIME, and Grad-CAM, to ensure transparency in clinical decision-making.

K5 – Engineering Design

- Design of a modular multi-phase pipeline integrating data acquisition, imaging analysis, and predictive modeling layers.
- Development of a hybrid fusion architecture that concatenates high-dimensional latent embeddings from CNNs with normalized clinical and wearable feature vectors.

K6 – Engineering Practice

- Practical implementation using an Intel Core i9 CPU, 32 GB RAM, and NVIDIA RTX 4090 GPU with the Adam optimizer and early stopping protocols.
- Application of image enhancement techniques such as Contrast-Limited Adaptive Histogram Equalization (CLAHE) and speckle noise reduction filters..

K7 – Engineering Comprehension

- Ability to analyze system behavior under noisy imaging conditions and interpret performance metrics such as Accuracy, Precision, Recall, and AUC-ROC.
- Interpretation of risk curves and progression heatmaps to assist clinicians in identifying high-risk patients transitioning to advanced disease stages.

K8 – Research Literature and Emerging Technologies

- Use of contemporary research in **privacy-preserving medical analytics**, deep learning for vascular segmentation, and wearable-based behavioral tracking.
- Evaluation and adaptation of recent **IEEE research findings** and clinical classification standards (CEAP) to design innovative, non-standard diagnostic solutions.

30/3/26

Signature

(Supervisor Name)

Dr. R. ANAANDH/ASP